1 Dimension of a Subspace

Theorem 1.1 (Dimension of a subspace)**.** *Let W be a subspace of a finite-dimensional vector space V. Then*

(a) W is finite-dimensional.

(b) $\dim W \leq \dim V$.

(c) There exists a subspace U of V such that $V = W \oplus U$.

(d) $W = V$ if and only if $\dim W = \dim V$.

Exercise 1.2. Prove Theorem 1.1.

Example 1.3. Let W be the subspace of $\mathbb{R}^5$ defined by

$$W = \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{R}^5 : x_1 = 3x_2 \text{ and } x_3 = 7x_4\}.$$

To find a basis of U, first note that

$$(3x_2, x_2, 7x_4, x_4, x_5) = x_2(3,1,0,0,0) + x_4(0,0,7,1,0) + x_5(0,0,0,0,1)$$

shows that W is spanned by $\{(3,1,0,0,0),(0,0,7,1,0),(0,0,0,0,1)\}$. Since

$$(0,0,0,0,0) = \alpha_1(3,1,0,0,0) + \alpha_2(0,0,7,1,0) + \alpha_3(0,0,0,0,1) = (3\alpha_1, \alpha_1, 7\alpha_2, \alpha_2, \alpha_3)$$

implies $\alpha_1 = \alpha_2 = \alpha_3 = 0$, the set $\{(3,1,0,0,0),(0,0,7,1,0),(0,0,0,0,1)\}$ is linearly independent and therefore a basis of W.

By the Plus/Minus Lemma, a linearly independent set can be extended to a basis of $\mathbb{R}^5$. Note that the vector $(1,0,0,0,0)$ is not in the span of $\{(3,1,0,0,0),(0,0,7,1,0),(0,0,0,0,1)\}$, since the first component is not 3 times the second. It follow from the Plus / Minus Lemma that

$$\{(3,1,0,0,0),(0,0,7,1,0),(0,0,0,0,1),(1,0,0,0,0)\}$$

is linearly independent. Similarly, vector $(0,0,1,0,0)$ is also not in the span of these vectors, since the third component is not 7 times the 4th component. It follows again from the Plus / Minus Lemma that

$$\{(3,1,0,0,0),(0,0,7,1,0),(0,0,0,0,1),(1,0,0,0,0),(0,0,1,0,0)\}$$

is linearly independent. Since $\mathbb{R}^5$ is 5-dimensional, this is a basis of $\mathbb{R}^5$. Finally, note that the proof of Theorem 1.1 shows that that $\mathbb{R}^5 = W \oplus U$, where the subspace U of $\mathbb{R}^5$ is given by

$$U = \text{span}\{(1,0,0,0,0),(0,0,1,0,0)\}$$

Exercise 1.4. Let $W = \{p \in \mathscr{P}_4(F) : p(6) = 0\}$.

(a) Find a basis of W.

(b) Extend the basis in (a) to a basis of $\mathscr{P}_4(F)$.

(c) Find a subspace U of $\mathscr{P}_4(F)$ such that $\mathscr{P}_4(F) = W \oplus U$.

Exercise 1.5. Suppose W and U are subspaces of V such that $V = W \oplus U$. Suppose also that $\{w_1, \ldots, w_m\}$ is a basis of W and $\{u_1, \ldots, u_n\}$ is a basis of U. Prove that $\{w_1, \ldots, w_m, u_1, \ldots, u_n\}$ is a basis of V.

Example 1.6. We will prove that the subspaces of $\mathbb{R}^2$ are precisely $\{0\}$, all lines in $\mathbb{R}^2$ containing the origin, and $\mathbb{R}^2$.

Since $\dim \mathbb{R}^2 = 2$, by part (b) of Theorem 1.1 any subspace must have dimension $0, 1$, or 2.

- The only 0-dimensional subspace is the trivial subspace $\{0\}$.
- Suppose W is a 1-dimensional subspace. Then W is the span of a single non-zero vector (x_1, x_2). That is, $(v_1, v_2) \in W$ if and only if there exists $\alpha \in \mathbb{R}$ such that $(v_1, v_2) = \alpha(x_1, x_2) = (\alpha x_1, \alpha x_2)$. This implies

  $$v_1 = \alpha x_1, \quad v_2 = \alpha x_2.$$

  These are the parametric equations of the line through $(0,0)$ and (x_1, x_2).
- If $\dim W = 2$, then $W = V$ by part (d) of Theorem 1.1.

Exercise 1.7. Prove that the subspaces of $\mathbb{R}^3$ are precisely $\{0\}$, all lines in $\mathbb{R}^3$ containing the origin, all planes in $\mathbb{R}^3$ containing the origin, and $\mathbb{R}^3$.

2 Dimension of a Sum

The next theorem is analogous to the formula

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2|$$

for the number of elements in the union of finite sets.

Theorem 2.1 (Dimension of a Sum)**.** *If V_1 and V_2 are subspaces of a finite-dimensional vector space, then*

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

Exercise 2.2. Fill in the steps in the proof of Theorem 2.1 below.

Proof. Let $\{v_1, \ldots, v_m\}$ be a basis of $V_1 \cap V_2$. Then $\dim(V_1 \cap V_2) = m$. Since $\{v_1, \ldots, v_m\}$ is a basis of $V_1 \cap V_2 \subseteq V_1$, it is linearly independent in V_1. Hence this list can be extended to a basis $\{v_1, \ldots, v_m, u_1, \ldots, u_j\}$ of V_1 by the Plus / Minus Lemma. Thus $\dim V_1 = m + j$. Similarly, we can extend $\{v_1, \ldots, v_m\}$ to a basis $\{v_1, \ldots, v_m, w_1, \ldots, w_k\}$ of V_2; thus $\dim V_2 = m + k$.

We will show that

$$\{v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k\} \tag{2.3}$$

is a basis of $V_1 + V_2$.

(a) Explain why this will complete the proof.

(b) Show that $V_1 + V_2 = \text{span}\{v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k\}$.

(c) Complete the proof by showing that $\{v_1, \ldots, v_m, u_1, \ldots, u_j, w_1, \ldots, w_k\}$ is linearly independent.

□

Corollary 2.4. *If V_1 and V_2 are subspaces of a vector space such that $V = V_1 \oplus V_2$, then*

$$\dim V = \dim V_1 + \dim V_2.$$

Exercise 2.5. Suppose U and W are both five-dimensional subspaces of $\mathbb{R}^9$. Prove that $U \cap W \neq \{0\}$.